

01 Introduction

Autistic children often struggle with reading comprehension even when decoding is strong. Two factors drive this:

- Narrative:** Sequencing events, causal links, characters' mental states
- Vocabulary:** Word meanings, difficulty level, word-concept links

MOSAIC (Multimodal Organizer for Story Analysis, Imagery, and Comprehension) is an authoring tool that turns stories into structured comics, targeting each driver: a genre-driven narrative structure makes story elements explicit, and word-level analysis lets learners simplify difficult vocabulary in place.

02 System Design



Figure 1: MOSAIC turns a story into a structured comic. (A) Prompt, genre, length. (B) Word Analysis scores vocabulary (AoA, Concreteness) with synonyms; Story Breakdown maps sections to Cohn's roles. (C) Characters extracted and rendered, editable by prompt or photo. (D) Sections render as panels with verbatim text. (E) Students reuse the structure to write their own.

1. Narrative Structure:

Organizes the story with Cohn's five narrative categories
Establisher → Initial → Prolongation → Peak → Release.
Writes one segment per stage (GPT-4o mini).

2. Word Analysis: Rates each word by age-of-acquisition (AoA) and Concreteness. Suggests lower-AoA synonyms suited to the audience.

3. Character Extraction: Extracts character descriptions (Claude Sonnet 4.6) and renders them in a consistent style (Gemini 3.1 Flash). Editable by prompt or photo upload.

4. Panel Generation: Writes scene directions from each narrative role and rendered with verbatim story text.

03 Application of Mosaic

FROM INFORMAL FEEDBACK

Teachers, parents, and autistic students surfaced three use cases during development.

Reading Comprehension + Composition

Narrative structure scaffolds comprehension; comics drive engagement.

→ An autistic learner, once reluctant to read, began composing her own stories after seeing a story's narrative structure.

Social Preparation

Comics rehearse new situations, easing the impact of novelty.

→ A parent used MOSAIC to teach behaviors and let their child engage with scenarios like a first day of school in advance.

Creative Agency

A shared scaffold supports composition; open endings preserve creative choice.

→ A teacher envisioned a create-your-own-adventure activity where he sets the opening; students create their own paths and endings.

04 Conclusion and Future Work

In conclusion, **MOSAIC** turns a story into a comic through genre-driven narrative structure and word-level vocabulary analysis, scaffolding the two factors behind reading-comprehension difficulty in autistic children.

For future work, we will conduct a study comparing text-only and comic conditions, and whether students compose more readily. Privacy safeguards, including content filtering, archetypal characters and data protections are underway.

05 References

- Cohn, N. (2013). Visual Narrative Structure. Cognitive Science.
- Blum et al. (2026). Disentangling local vs. global processing in autistic cognition: comics vs. text. Discourse Processes.